

THERMODYNAMICS

Basic Terminologies:

- ① System - part of the universe under thermodynamics investigation.
 - Open - Can exchange both energy and matter (coffee in open glass).
 - Close - Can exchange only energy (coffee in closed vessel).
 - Isolated - Cannot exchange matter and energy (Coffee in thermoflash).
- ② Surrounding - Anything outside the system.
- ③ Universe = System + Surrounding.
- ④ Boundary = Separate system and surrounding.
 - Rigid → Immovable
 - Non-Rigid → Movable
 - Adiabatic → $q = 0$
 - Diathermic → $q \neq 0$

Properties of System

Extensive	Intensive
Depend on amount of substance in system, and size of system. Volume, Gibbs, Internal energy, moles, enthalpy, surface area, mass, entropy, force, heat capacity.	Independent of amount of substance present in system and size of system. Molar volume, Density, Refractive index, Pressure, Temp, Surface tension, B.P., Freezing point, Vapour pressure.

$\rightarrow \frac{A}{B} \rightarrow \text{extensive} \quad \Delta \frac{A}{B} \rightarrow \text{Intensive}$
 e.g. $\rightarrow D = \frac{M}{V}, [C] = \frac{n}{V}$

State Function → Depend on initial and final state of system.
→ Do not depend on path followed by system.

e.g. → Enthalpy, Internal energy, Entropy, Gibbs Energy, Pressure, Temp, Volume.

Path function → Depend on path followed by system.
e.g. → Heat, Work.

Heat → Mode of energy transfer between system and surrounding.

→ Heat flowing into the system / Heat absorbed by system = $+q$

→ Heat flowing out of the system / Heat released by system = $-q$

Work → Mode of energy transfer between system and surrounding.

Non P.V work - Change in volume is not essential (e.g. electric work).

P.V work - Change in volume is essential.

Irreversible

$$W = -P_{\text{ext}} \Delta V$$

Reversible

$$W = -\int P \, dV$$

Work done by System → Expansion (-ve)

Work done on System → Compression (+ve)

Zeroth Law of Thermodynamics →

Two thermodynamic system are in thermal equilibrium with third system, then they are in thermal equilibrium with each other.

$$A \rightleftharpoons B, B \rightleftharpoons C, \text{ then } A \rightleftharpoons C$$

First Law of Thermodynamics →

→ Based on energy conservation principle.

→ Total energy of universe remains constant.

$$\Delta U = q + w \quad (\text{Internal energy} = \text{Heat} + \text{Work})$$